AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-11-02 | Entry window 2 November 2026 (full day) | 19 records screened in scope

19 RECORDS IN SCOPE (57 EXCLUDED)

8 OF 10 SUBTOPIC BANDS POPULATED

4 EFFECT ESTIMATES WITH A CI

2 TIER-A RECORDS

3 RECORDS CARRIED ON METADATA ALONE

Signal of the day

Toronto: the published four-station community AQHI missed other city sites by more than 2 AQHI units in both directions, across eight continuous sites in six weeks.

SWAPIT, January–March 2024. NO_2 dominated short-term AQHI risk and rose under calm winds; O_3 rose with wind speed; $PM_{2.5}$ carried both local and regional features. Against the NAPS record from 1997, median AQHI has fallen almost twice as fast in summer as in winter, leaving winter higher since 2013. **Where it is weak:** one six-week winter at eight sites, no CI, and the abstract gives neither instrument class nor site typology, so the deviation cannot be split into siting and method. **Why it matters here:** it puts a number on the gap between a sparse index and the intra-urban field. NO_2 rather than $PM_{2.5}$ drives the index here, so the sensor case rests on the multi-pollutant node, not on an optical PM node alone.

What changed today

- **Aged smoke is not fresh smoke.** Chamber aging raised ROS and oxidative DNA damage in BEAS-2B cells, most for smoldering emissions, while attenuating inflammatory signalling; cytotoxicity barely moved. No fold-changes in the abstract, and extracts on a submerged line, so dose relevance is open.
- **Low-cost optical sensors meeting vape aerosol, twice.** AirBeam3 in a bar: $PM_{2.5}$ mean 28.8, peak 84.7 $\mu g\,m^{-3}$ (pilot, no reference). Vendor-led school deployment, 37 sites, 18 months: over 300 incidents, no confusion matrix, alert fatigue and a smoke false positive reported. BACKFILL for the second, entry 4 May.
- **Inductive ferrous-aerosol sensing.** Micro-3D-printed impactor with liquid-metal coils: cut-offs 5.28 and 0.99 μm, nRMSE 6.72–8.95 %. Laboratory aerosol only; no humidity, drift or unit-to-unit data.
- **Hygroscopicity shifts with chemistry.** CMAQ, US summers 2013–2022: isoprene organosulfates move κ by −5.6 % net. A spatially structured term that a single humidity correction cannot carry.
- **Joint burden accounting.** 825 European cities: 213,889 deaths (89,224–344,223) jointly attributable to $PM_{2.5}$ and NO_2 in 2018, about a quarter below the sum of the single-pollutant figures. Abstract states a 76,833 gap; its printed numbers give 75,504. BACKFILL, Crossref 26 October.
- **Reproductive signals on thin exposure.** PMOS PR 1.22 (1.09–1.35) per SD $PM_{2.5}$, IDW from three stations; offspring hernia or hydrocele OR 1.07 (1.01–1.14) per $\mu g\,m^{-3}$, 53 events (BACKFILL, entry 4 May); a pregnancy-loss classifier with AUC 0.935 on internal validation only. All three are hypothesis-grade.
- **Exposure basis (x1).** Of 19 records, 2 rest on low-cost sensors (AirBeam3 bar pilot; school vape nodes), neither with a reference co-location in the abstract, and 4 on reference monitors. Health side with a recoverable basis (n=8): 50 % measured, 25 % modelled, 25 % proxy, but the measured share includes two station-interpolated residential estimates (three-station IDW; kriging), not personal measurement.

Window continuity

The 1 November issue closed with `last_entry_date` at 1 November. This issue was built on the **4 November 00:2x UTC (3 November 18:2x CST) run**; the run fired 19 hours after its 2 November 23:04 CST slot, and before 22:00 local, so 3 November is not opened and is left to the next run. It covers **2 November in full**. Three BACKFILL records are carried, the working cap, all marked with entry or deposit date. No rollup is owed: W45 (1–7 November) falls on Saturday 7 November.

Corpus analytics

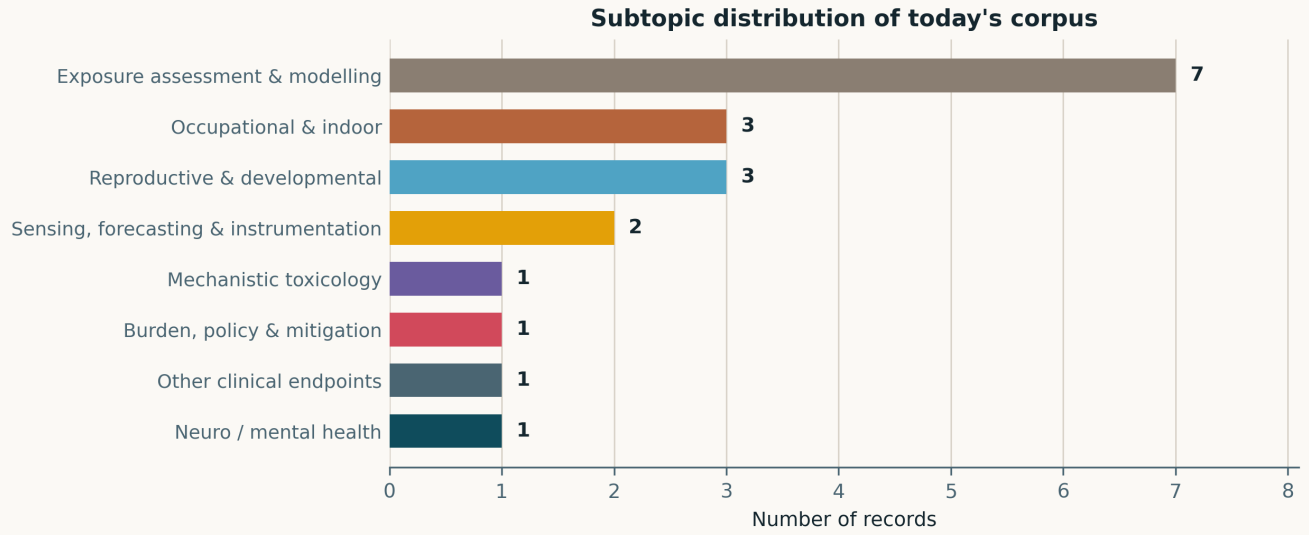


Fig. Subtopic assignment of the 19 records across eight bands: exposure assessment seven, reproductive three, occupational and indoor three, sensing two, and one each mechanistic, neuro, burden and other clinical. Cardiovascular and respiratory are empty today.

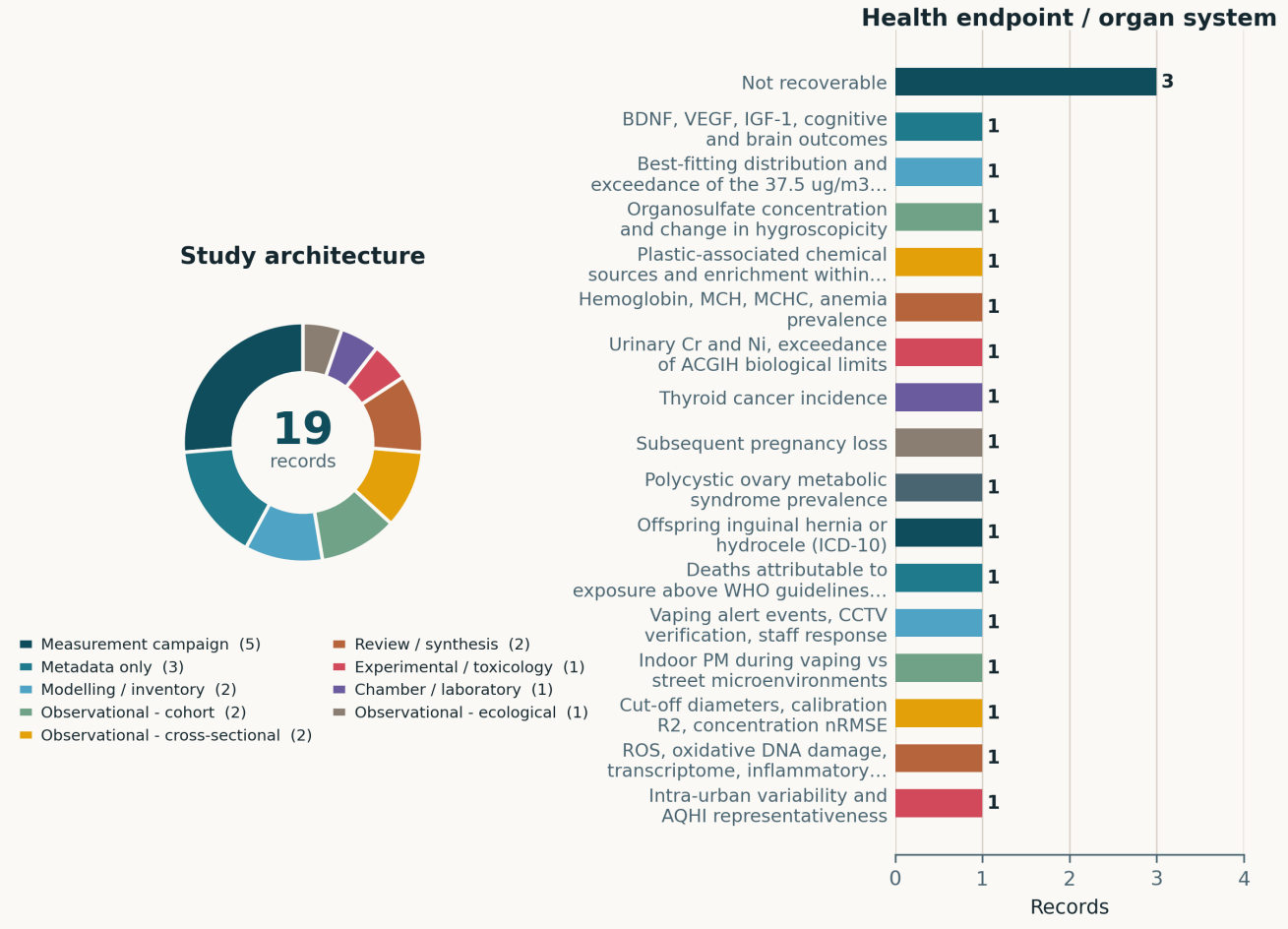


Fig. Left — architecture: measurement campaign five, metadata-only three, then two each cohort, cross-sectional, modelling and review, one each experimental, chamber or laboratory and ecological. Right — endpoints are near-singletons, as expected on a day led by exposure records.

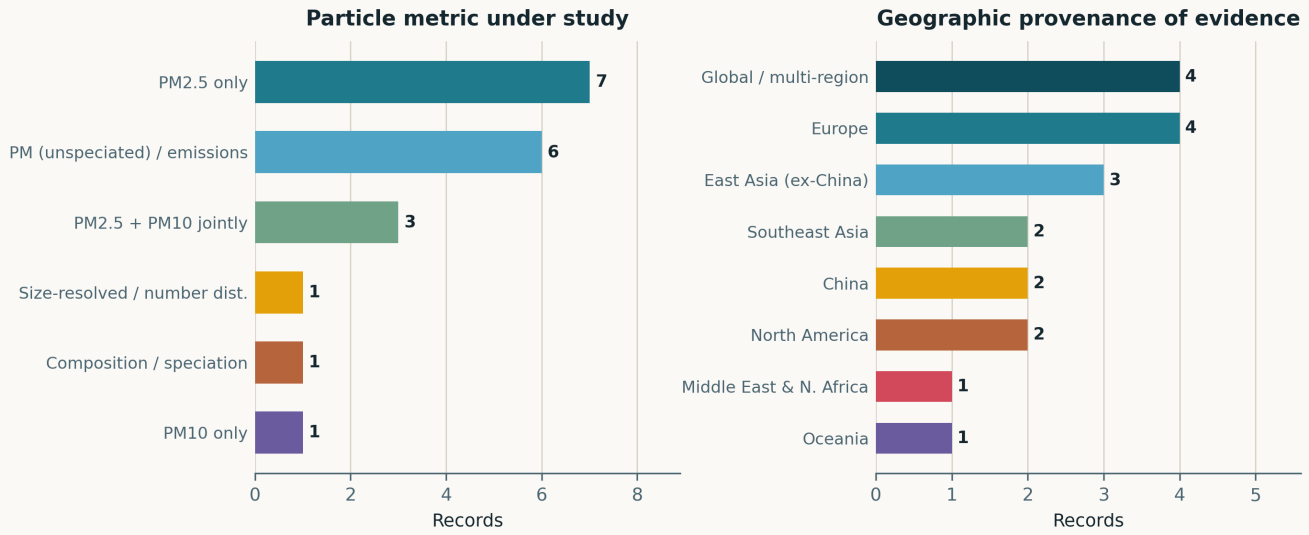


Fig. Left — particle metric: $PM_{2.5}$-only 7, unspeciated 6, $PM_{2.5}$+PM_{10} jointly 3, and one each PM_{10}-only, composition and size-resolved. Right — geography: Europe 4, global/multi-region 4 (two reviews, two with no site stated), East Asia ex-China 3, then China, North America and Southeast Asia 2 each, Oceania and Middle East 1 each.

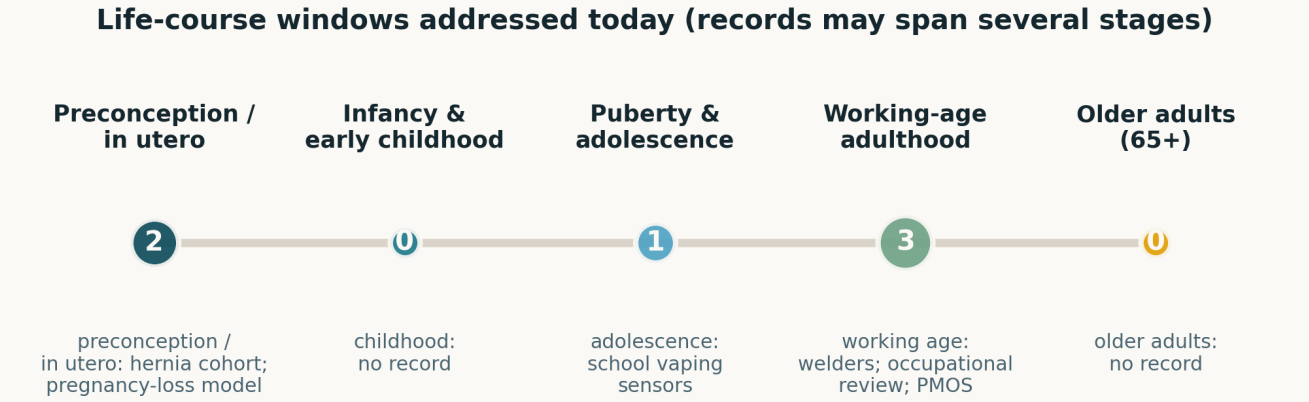


Fig. Life-course windows: two in utero or preconception (hernia cohort, pregnancy-loss model), one adolescent (school vaping sensors), three working-age (welders, occupational review, PMOS). No childhood or older-adult record.

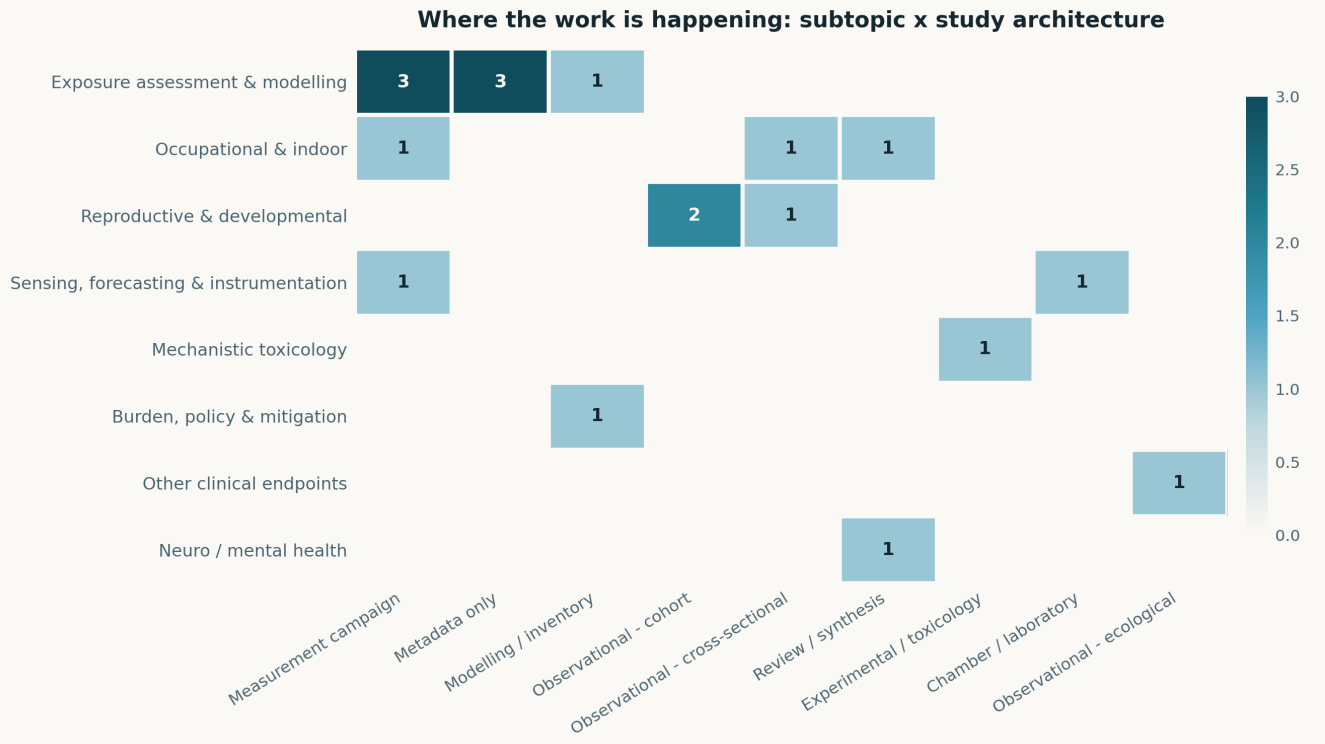


Fig. Subtopic × study architecture. Exposure assessment spreads across measurement, modelling and metadata-only; three of its seven records are metadata-only or discussion. Sensing sits in measurement and laboratory cells only, with no co-location design today.

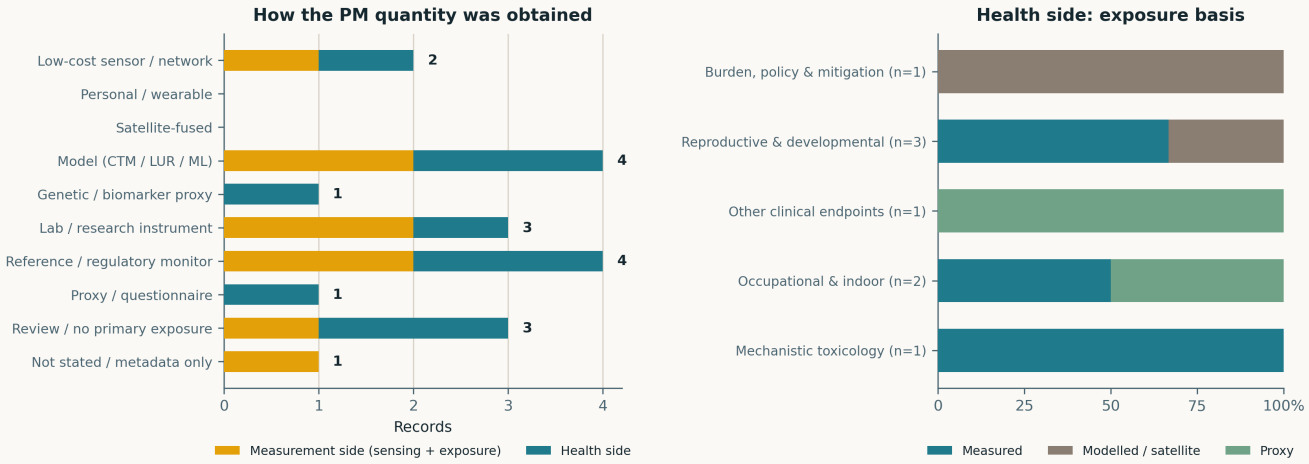


Fig. How each record obtained its PM quantity, classified from the exposure field. 2 of 19 records rest on a low-cost sensor or sensor network; 4 on reference or regulatory monitors. Health side, where an exposure basis is recoverable (n=8): measured 50 %, modelled or satellite-fused 25 %, proxy 25 %. Right panel sorted by modelled share: the subtopics at the bottom are where a measured low-cost network exposure would add most.

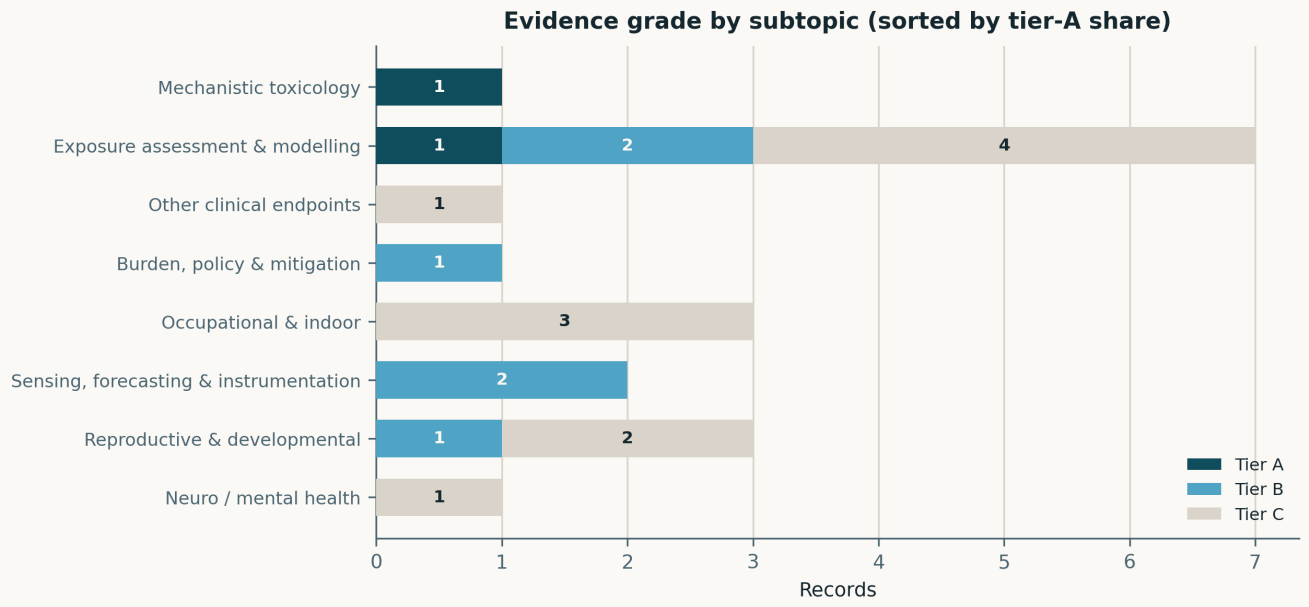


Fig. Tier mix per subtopic. 2 of 19 records are tier A (11 %). Highest tier-A share: *Mechanistic toxicology* (1 of 1).

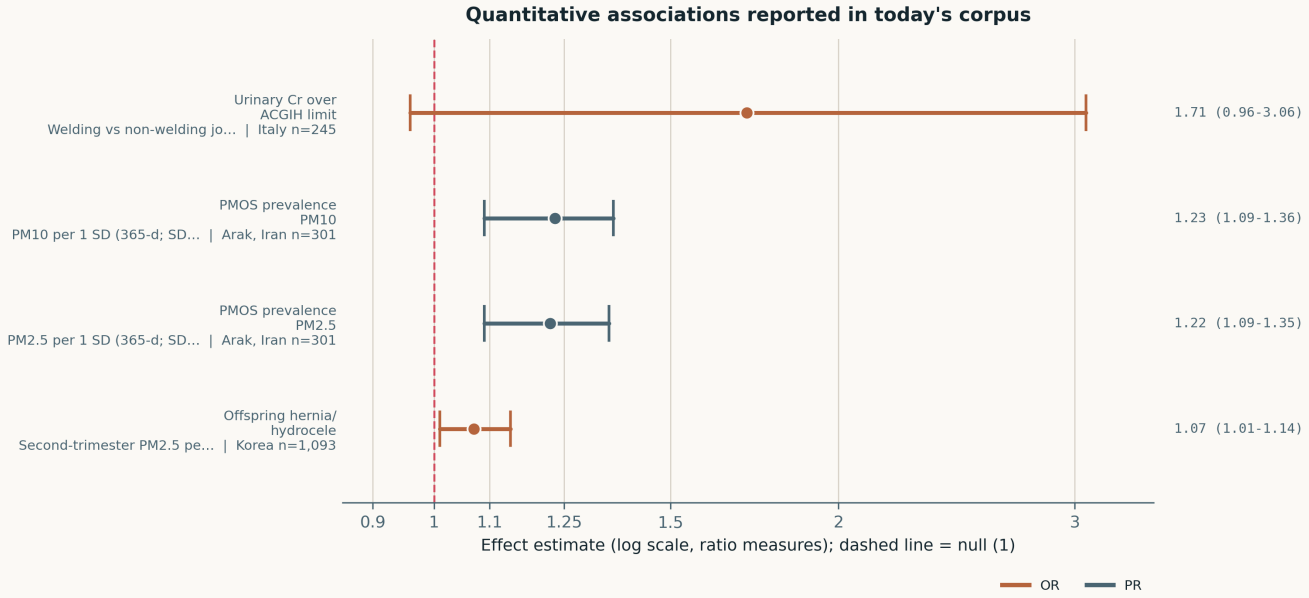


Fig. Ratio scale, four estimates from three studies. Increments differ (per SD with undisclosed width in Arak; per 1 $\mu g\,m^{-3}$ in Korea; a job-role contrast for welders), and the two Arak estimates come from the same 301 women: nothing here is comparable across studies or poolable. The Korean carcinogen-emission RR is not a PM estimate and is not drawn.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

Cho exercise under particulate matter exposure BDNF neurotrophic narrative review 2026 Atmosphere

Exercise Under Particulate Matter Exposure: Neurotrophic Responses and Functional Brain Outcomes

Focused narrative review, PubMed and Web of Science, 37 reports (33 main, four indirect). Circulating BDNF responses to exercise under PM were heterogeneous; VEGF and IGF-1 gave complementary peripheral evidence. Human studies mixed attenuation in selected cognitive, dementia and imaging contrasts with retained activity benefit and possible buffering. Central molecular evidence from inhalation was limited; some animal improvements came from intravenous or instillation models. The authors find support for neither a uniform loss of exercise benefit nor a general reversal threshold. No pooling. Limitation: non-systematic, overlapping cohorts possible. Relevance: low-moderate - the stated gap is activity-specific exposure, which is what wearable PM sensing alongside exercise intensity could supply.

doi: 10.3390/atmos17111077

Reproductive & developmental 3 records

Kim prenatal $PM_{2.5}$ pediatric inguinal hernia hydrocele kriging cohort 2026 J Clin Med

Prenatal Fine Particulate Matter ($PM_{2.5}$) Exposure and the Risk of Pediatric Inguinal Hernia or Hydrocele: A Retrospective Cohort Study

BACKFILL (entry 4 May 2026). 1,093 mother-offspring pairs, one tertiary centre, deliveries July 2016-June 2019, follow-up to March 2024; 53 cases (4.85%). Second-trimester $PM_{2.5}$ OR 1.07 (1.01-1.14) per 1 $\mu g\,m^{-3}$; an IQR increase gave OR 1.64 (no CI, IQR width not stated). Male sex OR 24.71 (5.95-102.54). AUC 0.804. Limitation: 53 events against several trimester windows with no stated multiplicity correction, referral-centre cohort, hernia and hydrocele pooled, and a mechanism inferred rather than tested. Treat as hypothesis. Relevance: low - kriging from fixed stations, with sparse-network error in a trimester-specific contrast, the setting where denser residential estimates would matter.

doi: 10.3390/jcm15083089

Karimi PMOS polycystic ovary metabolic syndrome Arak Iran $PM_{2.5}$ PR IDW 2026 Chemosphere

Air pollution and polycystic ovary metabolic syndrome (PMOS) in a heavily industrialized Middle Eastern city: A cross-sectional study.

301 women at gynecology clinics in Arak (153 PMOS, 148 non-PMOS), April 2024-March 2025. Modified Poisson, per 1-SD: $PM_{2.5}$ PR 1.22 (1.09-1.35), PM_{10} PR 1.23 (1.09-1.36), NO_2 1.21 (1.07-1.33), SO_2 1.15 (1.01-1.28). Cases had 3.00 $\mu g\,m^{-3}$ more $PM_{2.5}$. Sensitivity windows and propensity matching agreed. SD widths are not in the abstract. Limitation: exposure from three stations by IDW in an industrial city carries little real spatial contrast, so residence and socioeconomic position are confounded with it; clinic controls and prevalent cases rule out temporal inference. Relevance: moderate - a study whose result hangs on three monitors is where a small low-cost network would directly test the contrast.

doi: 10.1016/j.chemosphere.2026.145136

Qiu stage-specific models pregnancy loss recurrent abortion $PM_{2.5}$ O_3 random forest 2026 Int J Womens Health

Stage-Specific Models for Predicting Subsequent Pregnancy Loss in Women with Recurrent Spontaneous Abortion.

Single-centre prospective cohort of 470 women with recurrent spontaneous abortion, 86 (18.3%) with another loss. Three staged models by logistic regression, random forest and XGBoost, repeated 10-fold cross-validation. Third-stage random forest ROC-AUC 0.935 (0.874-0.996); XGBoost 0.870 (0.801-0.938), after adding gestational age, hormones and $PM_{2.5}$ and O_3 metrics. No PM effect estimate. Limitation: internal validation only, 86 events or fewer against many predictors, and gestational age at measurement risks leaking the outcome; the PM contribution cannot be separated from the bundle, and the exposure source is not in the abstract. Relevance: low - PM is a feature in a classifier, and this is no evidence that exposure drives discrimination.

doi: 10.2147/ijwh.s637498

Mechanistic toxicology 1 record

Stegmann atmospheric aging biomass burning particle toxicity BEAS-2B PHOTO-LAC 2026 Environ Sci Technol

Atmospheric Aging Modifies the Toxicological Profile of Biomass Burning Particles in Epithelial Lung Cells.

Biomass burning particles from controlled flaming and smoldering combustion were aged in the PHOTO-LAC chamber under alternating photochemical and dark conditions, then extracts dosed on BEAS-2B bronchial cells. Aging enhanced intracellular ROS and oxidative DNA damage, most for smoldering emissions, with cytotoxicity largely unchanged. Inflammatory and matrix-remodelling responses were attenuated with aging in smoldering particles and stable in flaming particles. No fold-changes or doses in the abstract. Limitation: extracts on a submerged immortalised line, not air-liquid interface deposition, and one chamber aging protocol, so dose relevance to ambient plumes is open. Relevance: moderate - mass from a sensor is blind to aging state; a plume reading the same $\mu g\,m^{-3}$ near source and downwind need not be the same exposure.

doi: 10.1021/acs.est.6c02821

Exposure assessment & modelling 7 records

Kutarna SWAPIT Toronto winter NO_2 O_3 $PM_{2.5}$ AQHI representativeness 2026 J Air Waste Manag Assoc

Insights into urban concentrations and variability of $NO(2)$, $O(3)$, and $PM(2.5)$ from the study of winter air pollution in Toronto (SWAPIT).

Eight continuous sites across Toronto, six weeks January-March 2024, plus NAPS records from 1997. The published community AQHI, built from four NAPS stations, deviated by more than 2 AQHI units above and below values at other sites. NO_2 dominated short-term AQHI risk and rose under calm winds; O_3 rose with wind speed; $PM_{2.5}$ showed local and regional features. Median AQHI fell almost twice as fast in summer as in winter, leaving winter higher since 2013. No CI. Limitation: one six-week winter at eight sites, and the abstract names neither instrument class nor site typology, so the deviation cannot be assigned to siting or to method. Relevance: high - a quantified gap between a four-station index and the intra-urban field, the case for denser sensing, though NO_2 rather than $PM_{2.5}$ drives it.

doi: 10.1080/10962247.2026.2743096

Wu airborne plastics plastic-associated chemicals $PM_{2.5}$ Singapore PMF I168O 2026 Environ Sci Technol

Fingerprints of Airborne Plastics in Urban $PM_{2.5}$: Applications and Implications

240 daily $PM_{2.5}$ samples, Singapore 2018-2020, nine plastic-associated chemicals, seven apportioned sources. I168O made up 90% of quantified chemicals and identified at least 40% more $PM_{2.5}$ linked to respirable airborne plastic chemicals. A resolved plastic production, use and burning factor held 37% of quantified chemicals; traffic and road dust held over 20% of near-range chemicals, with DNBP/DEHP above 1 proposed as a non-exhaust marker. Over 90% of DEP sat with transported peat smoke. No uncertainty on factor shares in the abstract. Limitation: one site is implied, and I168O is an additive oxidation product with other potential sources, so the factor measures additives, not plastic mass. Relevance: low for mass sensing - a speciation layer invisible to optical nodes.

doi: 10.1021/acs.est.6c09617

Gonzalez organosulfates CMAQ hygroscopicity kappa US summers 2013-2022 2026 Environ Sci Technol

Modeling Decadal Organosulfates in Particulate Matter to Improve Estimations of Aerosol Hygroscopicity in the United States

CMAQ with updated isoprene organosulfate chemistry, contiguous US, summers 2013-2022, evaluated on IMPROVE organosulfur and sulfate plus speciated filters, with an interpretable ML surrogate. Summer mean organosulfate up to 0.51 $\mu g\,m^{-3}$ in the east. Net change in hygroscopicity kappa -5.6% (direct +2.8%, indirect -8.4% through sulfate reduction). No CI on these. Limitation: summer only, isoprene-derived only, and validation rests on sparse speciated measurements; evaluation metrics are not in the abstract. Relevance: moderate - humidity corrections for optical sensors commonly assume a kappa; a chemistry-driven shift of this size, spatially structured, is one physical reason a single continental correction does not transfer.

doi: 10.1021/acs.est.6c08752

Jiratampradab daily $PM_{2.5}$ probability distribution upper northern Thailand log-normal 2026 Atmosphere

Probability Distribution Analysis of Daily $PM_{2.5}$ Concentration at Five Stations in Upper Northern Thailand

1,096 daily values at each of Chiang Rai, Mae Hong Son, Chiang Mai, Nan and Lampang, June 2023-May 2026. Log-normal, gamma and Weibull fitted by maximum likelihood and compared by Kolmogorov-Smirnov, Anderson-Darling, AIC and BIC. Log-normal won at all stations. Exceedance of the 24-h 37.5 $\mu g\,m^{-3}$ standard: Nan 25.12%, Lampang 21.70%, Chiang Rai 20.87%. No CI. Limitation: treating strongly autocorrelated, bimodal-season data as independent draws invalidates the goodness-of-fit p-values and understates tail uncertainty, and three years do not constrain extremes. Relevance: low - a descriptive baseline for a haze region where low-cost networks are already dense; useful for setting sensor range and alert thresholds.

doi: 10.3390/atmos17111081

Wayland discussion data fusion air pollution exposure modeling metadata only 2026 J Air Waste Manag Assoc

Discussion of "Data fusion for air pollution exposure modeling".

METADATA ONLY. No abstract available from PubMed or Crossref at issue time, and no finding is asserted. PubMed lists this as a discussion at J Air Waste Manag Assoc 76(11):908-916, with authors from the retired EPA, Resources for the Future, Emory and Tennessee among others. Limitation: content unread. Relevance: on title alone high - data fusion is where sensor, monitor, satellite and model fields are combined; retrieve the full text and the paper under discussion.

doi: 10.1080/10962247.2026.2737061

Sharma $PM_{2.5}$-bound water soluble organic carbon urban sources metadata only 2026 Atmos Pollut Res

Variability in sources and processes of $PM_{2.5}$-bound Water Soluble Organic Carbon over an urban location in the North-Western Himalayas

METADATA ONLY. Elsevier day-0 deposit, no abstract in Crossref or Semantic Scholar at issue time; no finding is asserted. The title places it in $PM_{2.5}$ water-soluble organic carbon source and process variability at an urban location. Limitation: content unread. Relevance: unknown, retry the abstract at the weekly sweep.

doi: 10.1016/j.apr.2026.103262

Wei dust storm events Northeast China GEOS-Chem PM measurements metadata only 2026 Atmos Pollut Res

Dust storm events in Northeast China by GEOS-Chem modeling and PM measurements

METADATA ONLY. Elsevier day-0 deposit, no abstract in Crossref or Semantic Scholar at issue time; no finding is asserted. The title places it in dust storm events over Northeast China using GEOS-Chem and PM measurements. Limitation: content unread. Relevance: unknown on content; dust regimes are where low-cost optical sensors undercount, so retry the abstract at the weekly sweep.

doi: 10.1016/j.apr.2026.103259

Sensing, forecasting & instrumentation 2 records

Park micro-3D-printed liquid-metal coil inductive ferrous aerosol impactor sensor 2026 Lab Chip

Micro-3D-printed integrated system with liquid-metal coils for simultaneous size-selective sampling and concentration sensing of respirable ferrous metal aerosols.

Monolithic micro-3D-printed platform: two-stage inertial impactor (aerodynamic cut-offs 5.28 and 0.99 um), aqueous particle transfer, 100 um sensing channels, embedded EGaIn solenoids (50 turns, 907.1 nH, maximum Q 9.1 at 30 MHz). Fe-only aerosol: peak-rate calibration R2 0.900 and 0.910, time-resolved nRMSE 6.72% and 8.95% for the two fractions. Arizona test dust background: R2 0.9428 and 0.9348 against APS total particle increments. Responses persisted over a 60-min mixed-aerosol challenge. Limitation: laboratory-generated aerosol only, no field, humidity, drift or unit-to-unit data, and the dust-background R2 is against total count, not ferrous mass. Relevance: moderate - composition-selective inductive sensing is an orthogonal channel to optical scattering, but far from a deployable node.

doi: 10.1039/d6lc00748a

Sharon IoT vape detection school $PM_{2.5}$ PM_{10} 37 locations 18 months 2026 Int J Environ Res Public Health

Smart Vape Detection in Schools for Mitigating Student E-Cigarette Use

BACKFILL (entry 4 May 2026). Low-cost IoT PM nodes at 37 restroom and change-room locations in one Australian independent school, one-minute data for 18 months, threshold alerts with school-led CCTV verification. More than 300 vaping-related incidents, clustered by time of day and among senior students; alerts showed rapid concurrent $PM_{2.5}$ and PM_{10} excursions. Staff response declined through alert fatigue, and a regional smoke event produced false positives. No sensitivity, specificity or denominator of unverified alerts in the abstract. Limitation: first-author affiliation is the IoT vendor; 'highly effective' is claimed without a confusion matrix. Relevance: high - event detection from PM transients on low-cost nodes, with the human-response failure and the need for context-aware logic documented.

doi: 10.3390/ijerph23040501

Occupational & indoor 3 records

Borgini e-cigarette real-life indoor PM AirBeam3 bar traffic walk pilot 2026 Multidiscip Respir Med

Investigating the Real-Life Indoor Air Quality Impact of E-Cigarette Vaping.

Pilot walk with an AirBeam3 along low- and heavy-traffic roads, then inside a bar with three people vaping. Bar $PM_{2.5}$ averaged 28.8 $\mu g\,m^{-3}$ with a peak of 84.7 $\mu g\,m^{-3}$, 6.5 times the low-traffic and 3.1 times the heavy-traffic street value. No CI; a single session is implied, with no replicate or vape-free control in the same room. Limitation: e-cigarette aerosol is volatile liquid droplets, outside an ambient-dust factory calibration, and no gravimetric or reference co-location is reported, so the mass values are nominal; bar sources other than vaping are uncontrolled. Relevance: direct - a low-cost optical sensor reporting a mass ratio in an aerosol regime it was never characterised for. Read it as a relative signal.

doi: 10.5826/mrm.2026.1117

Zulian urinary chromium nickel welders TIG metalworkers Friuli biomonitoring 2026 Occup Med (Lond)

Biological chromium and nickel monitoring in metalworkers: welders versus non-welders.

245 male metalworkers, Friuli Venezia Giulia (179 welders, 66 non-welders), end-of-week urine by ICP-MS. Welders had higher median Cr (0.75 vs 0.56 ug/l, P=0.020) and Ni (1.70 vs 1.10 ug/l, P=0.0001). 50.0% exceeded the Cr limit of 0.7 ug/l. Welding vs non-welding OR 1.71 (0.96-3.06) for exceeding the Cr limit. Age and smoking were unrelated. Limitation: no airborne fume measurement, so job title stands in for exposure, non-welders share the shop air, and the group contrast is correspondingly compressed; Cr speciation not reported. Relevance: low for ambient sensing - a biomarker reference for a workplace aerosol that task-level PM sensors could stratify.

doi: 10.1093/occmed/kqag126

Khoshakhlagh occupational airborne pollutants hematological outcomes systematic review 2026 J Appl Toxicol

Occupational Airborne Pollutant Exposure and Hematological Outcomes in Workplace Settings: A Systematic Review.

PRISMA review of 1,855 records over five decades, 55 studies, predominantly developing and industrialising countries, JBI appraisal, no pooled estimate. Workplace VOCs, heavy metals such as lead, and PM were linked to subclinical haematological change: lower hemoglobin, altered MCH and MCHC. Linear correlations with pollutant levels were often weak despite high concentrations. BMI associated positively with hemoglobin; age weakly, read as a healthy-worker effect. Limitation: no effect size, PM not separated from co-exposures, and a protective BMI 'buffer' is confounding by nutrition until shown otherwise. Relevance: low - weak exposure-response at high concentrations is also what uncalibrated or area-based workplace exposure metrics would produce.

doi: 10.1002/jat.70474

Burden, policy & mitigation 1 record

Dyer urban burden of disease 825 European cities joint $PM_{2.5}$ NO_2 green space 2026 Eur J Public Health

Urban burden of disease estimation for policy making: health impacts of environmental stressors

BACKFILL (Crossref deposit 26 October 2026; congress abstract). Comparative risk assessment across 825 European cities, three reporting periods 2015-2024. Exceeding WHO guidelines gave 213,889 deaths (95% CI 89,224-344,223) jointly attributable to $PM_{2.5}$ and NO_2 in 2018, reported as 76,833 fewer than summing single-pollutant estimates ($PM_{2.5}$ 145,747; NO_2 143,646; the printed figures actually differ by 75,504). Limitation: abstract only; the joint method is not specified, the interval is wide, and attribution inherits the concentration-response function and the exposure model with no measured validation stated. Relevance: moderate - summing pollutant burdens overstates by about a quarter here, and city-scale burden accounting is where dense sensor fields could test the modelled surface.

doi: 10.1093/eurpub/ckag169.841

Other clinical endpoints 1 record

Kim district carcinogen emissions thyroid cancer Korea Bayesian PM_{10} null 2026 J Prev Med Public Health

Association Between District-level Aggregate Carcinogen Emissions and Thyroid Cancer Incidence in Korea: An Ecological Study Using Bayesian Spatial Models.

229 Korean districts, thyroid cancer 2014-2018, Bayesian hierarchical Poisson with mutual adjustment. Aggregate IARC Group 1/2A/2B release indicator RR 1.04 (95% CrI 1.01-1.06) per SD overall, 1.04 (1.02-1.06) in women, 1.02 (0.99-1.06) in men. PM_{10}, NO_2, SO_2, radon and nuclear-plant proximity showed no association after urban-rural adjustment; CO was inverse (0.96, 0.93-0.99); population density 1.13 (1.06-1.21). Limitation: ecological, and thyroid cancer incidence in Korea is strongly shaped by screening and detection intensity; authors call it hypothesis-generating. PM_{10} estimate not reported in the abstract. Relevance: low - a coarse-metric null at district scale, informative only in showing that urbanicity adjustment absorbs the PM signal.

doi: 10.3961/jpmph.26.726

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Cho exercise under particulate matter exposure BDNF neurotrophic narrative review	Exercise Under Particulate Matter Exposure: Neurotrophic Responses and Functional Brain Outcomes	*Atmosphere*	Narrative review	Particulate or traffic-related pollution during exercise, heterogeneous metrics	Global / multi-region
Reproductive & developmental					
Karimi PMOS polycystic ovary metabolic syndrome Arak Iran $PM_{2.5}$ PR IDW	Air pollution and polycystic ovary metabolic syndrome (PMOS) in a heavily industrialized Middle Eastern city: A cross-sectional study.	*Chemosphere*	Cross-sectional study	365-day residential $PM_{2.5}$ and PM_{10} by inverse-distance weighting from three stations	Iran (Arak)
Kim prenatal $PM_{2.5}$ pediatric inguinal hernia hydrocele kriging cohort	Prenatal Fine Particulate Matter ($PM_{2.5}$) Exposure and the Risk of Pediatric Inguinal Hernia or Hydrocele: A Retrospective Cohort Study	*J Clin Med*	Retrospective cohort	Monthly residential $PM_{2.5}$ by kriging from fixed monitoring stations	South Korea
Qiu stage-specific models pregnancy loss recurrent abortion $PM_{2.5}$ O3 random forest	Stage-Specific Models for Predicting Subsequent Pregnancy Loss in Women with Recurrent Spontaneous Abortion.	*Int J Womens Health*	Prediction model development (prospective cohort)	Pregnancy-course ambient $PM_{2.5}$ and O3 metrics as predictors	China
Mechanistic toxicology					
Stegmann atmospheric aging biomass burning particle toxicity BEAS-2B PHOTO-LAC	Atmospheric Aging Modifies the Toxicological Profile of Biomass Burning Particles in Epithelial Lung Cells.	*Environ Sci Technol*	In vitro cell exposure	Fresh vs chamber-aged biomass burning particle extracts, flaming and smoldering	Germany (laboratory)
Occupational & indoor					
Borgini e-cigarette real-life indoor PM AirBeam3 bar traffic walk pilot	Investigating the Real-Life Indoor Air Quality Impact of E-Cigarette Vaping.	*Multidiscip Respir Med*	Pilot field measurement	PM1, $PM_{2.5}$, PM_{10} by portable optical particle counter (AirBeam3)	Italy
Khoshakhlagh occupational airborne pollutants hematological outcomes systematic review	Occupational Airborne Pollutant Exposure and Hematological Outcomes in Workplace Settings: A Systematic Review.	*J Appl Toxicol*	Systematic review	Occupational PM, VOCs and heavy metals across 55 studies	Global / multi-region
Zulian urinary chromium nickel welders TIG metalworkers Friuli biomonitoring	Biological chromium and nickel monitoring in metalworkers: welders versus non-welders.	*Occup Med (Lond)*	Cross-sectional study	Welding fume by job role; urinary Cr and Ni by ICP-MS (no airborne measure)	Italy
Sensing, forecasting & instrumentation					
Park micro-3D-printed liquid-metal coil inductive ferrous aerosol impactor sensor	Micro-3D-printed integrated system with liquid-metal coils for simultaneous size-selective sampling and concentration sensing of respirable ferrous metal aerosols.	*Lab Chip*	Instrument development and laboratory validation	Respirable ferrous aerosol, 1-5 and >5 um fractions, inductive detection	South Korea (laboratory)
Sharon IoT vape detection school $PM_{2.5}$ PM_{10} 37 locations 18 months	Smart Vape Detection in Schools for Mitigating Student E-Cigarette Use	*Int J Environ Res Public Health*	Longitudinal case study of a sensor deployment	$PM_{2.5}$ and PM_{10} at one-minute resolution, threshold alerting	Australia
Exposure assessment & modelling					
Gonzalez organosulfates CMAQ hygroscopicity kappa US summers 2013-2022	Modeling Decadal Organosulfates in Particulate Matter to Improve Estimations of Aerosol Hygroscopicity in the United States	*Environ Sci Technol*	Chemical-transport modelling	Isoprene-derived organosulfates and aerosol hygroscopicity from CMAQ, evaluated on IMPROVE	United States
Jiratampradab daily $PM_{2.5}$ probability distribution upper northern Thailand log-normal	Probability Distribution Analysis of Daily $PM_{2.5}$ Concentration at Five Stations in Upper Northern Thailand	*Atmosphere*	Statistical analysis of monitoring data	Daily $PM_{2.5}$ at five monitoring stations, June 2023-May 2026	Thailand (upper north)

First author	Focus	Journal	Design	Metric	Region
Kutarna SWAPIT Toronto winter NO2 O3 $PM_{2.5}$ AQHI representativeness	Insights into urban concentrations and variability of NO(2), O(3), and PM(2.5) from the study of winter air pollution in Toronto (SWAPIT).	*J Air Waste Manag Assoc*	Multi-site monitoring campaign	$PM_{2.5}$ with NO2 and O3 at eight continuous sites, plus NAPS history from 1997	Canada (Toronto)
Sharma $PM_{2.5}$-bound water soluble organic carbon urban sources metadata only	Variability in sources and processes of $PM_{2.5}$-bound Water Soluble Organic Carbon over an urban location in the North-Western Himalayas	*Atmos Pollut Res*	Metadata only: no abstract in Crossref at issue time	$PM_{2.5}$-bound water-soluble organic carbon (from title)	Not stated
Wayland discussion data fusion air pollution exposure modeling metadata only	Discussion of "Data fusion for air pollution exposure modeling".	*J Air Waste Manag Assoc*	Metadata only: invited discussion of a data-fusion paper	Not recoverable	Not stated
Wei dust storm events Northeast China GEOS-Chem PM measurements metadata only	Dust storm events in Northeast China by GEOS-Chem modeling and PM measurements	*Atmos Pollut Res*	Metadata only: no abstract in Crossref at issue time	PM from measurement and GEOS-Chem (from title)	China (Northeast)
Wu airborne plastics plastic-associated chemicals $PM_{2.5}$ Singapore PMF I168O	Fingerprints of Airborne Plastics in Urban $PM_{2.5}$: Applications and Implications	*Environ Sci Technol*	Source apportionment	Nine plastic-associated chemicals in 240 daily $PM_{2.5}$ filters, 2018-2020	Singapore
Burden, policy & mitigation					
Dyer urban burden of disease 825 European cities joint $PM_{2.5}$ NO2 green space	Urban burden of disease estimation for policy making: health impacts of environmental stressors	*Eur J Public Health*	Comparative risk assessment	Modelled $PM_{2.5}$ and NO2, joint exposure estimation, against WHO guidelines	Europe (825 cities)
Other clinical endpoints					
Kim district carcinogen emissions thyroid cancer Korea Bayesian PM_{10} null	Association Between District-level Aggregate Carcinogen Emissions and Thyroid Cancer Incidence in Korea: An Ecological Study Using Bayesian Spatial Models.	*J Prev Med Public Health*	Observational - ecological	District PM_{10} as a covariate; primary exposure a carcinogen release indicator	South Korea

Provenance and method

Entry window **2 November 2026**, single day (Monday). Harvester legs: **PubMed** health **9**, sensing **2**; **Europe PMC 9**, no AGRICOLA records; **Crossref by ISSN 54**; **arXiv** 150 kB, no in-window entry; **OpenAlex** 0 returned (the OpenAlex connector is not on this session). 69 unique, 1 already carried, 68 fresh; PubMed connector (EDAT 2 Nov, three axes) 16 PMIDs plus one Amass check, 4 appended (**72 screened**). Abstracts: 10 recovered by efetch; Elsevier day-0 deposits remain empty. **Amass** BiomedCore (ingest date filter) surfaced **3 backfill** records admitted (two from Amass, one already held from 1 November), 1 rejected and 1 held. **Consensus** calibration sweep: 10 hits, 0 new. bioRxiv/medRxiv and alphaXiv connectors are not on this session. **Scite** retraction check unavailable (connector requires a paid plan); Amass reports no retraction on the PMID-bearing backfills. **Wiley Scholar Gateway** not on this session. **19 admitted** (16 window, 3 backfill), **57 rejected** with logged reasons. **Trial watch** 2–4 November: ClinicalTrials.gov 1, NCT07914010 (REACT, UC Berkeley: PurpleAir indoor monitors with the indicator light randomised on or off, 100 households, co-primary indoor $PM_{2.5}$ and BCSS/ACT); Amass TrialCore 20 returned, 2 relevant (ACTRN12626001238369, Melbourne HEPA RCT in COPD, n=200; ChiCTR2600134305). **3 carried on metadata alone**. Abstracts remain the copyright of their publishers.